DATA STRATEGIST CASE STUDY · MYNT DATA OFFICE

Give Me The Next AAA Title

A data strategy for deciding which film a studio should fund next — from 750,000 IMDb titles to a greenlight score you can act on.

[Your Name]

Presented to the studio management team & data science leads

EXECUTIVE SUMMARY

Fund proven talent on an event concept, and let a score gate the slate

74%

Bet on talent first

The track records of the director, writer, top-billed cast and producers carry **74%** of the model's predictive weight — nearly 4× genre, runtime and franchise combined.

3.5×

Build an event concept

Adventure, Animation and Action appear 1.8–3.5× more often among 2020–25 hits than among all releases. Hits run ~120 minutes.

19×

Gate the slate with a score

On unseen 2017–2025 films, the top 2% of scored projects became AAA hits 61% of the time vs a 3.2% market base rate.

APPROACH

I framed the brief as one decision and four questions

1 Decision

Which project should we fund as our next AAA title?

Business framing with management

2 Success

What does a 'hit' mean in money terms?

Metric validated against box office

3 Prediction

What, known before greenlight, predicts it?

Leakage-safe ML, out-of-time test

4 Value

How will we know the model paid off?

Backtest + pilot design + KPIs

The brief's three exploration questions feed step 2; the model is step 3. **Step 4 is where most data projects stall**, so it gets its own section.

DATA FOUNDATION

A SQL warehouse on DuckDB, built to port straight to the cloud

Raw	7 IMDb dumps incl. title.principals, gzip, streamed — never unzipped
Silver	Typed SQL tables: films, ratings, crew, cast billing, producers, release footprint
Augment	Budget & gross (TMDB, The Numbers); Wikipedia 2020 billing as a cross-check
Gold	One row per film: the single source every chart and model reads

750,407
feature films in IMDb

60,352
analysis universe: 1970–2025, ≥500 votes

5,416
films with budget + worldwide gross

78%
of Wikipedia's 2020 billings matched by IMDb principals

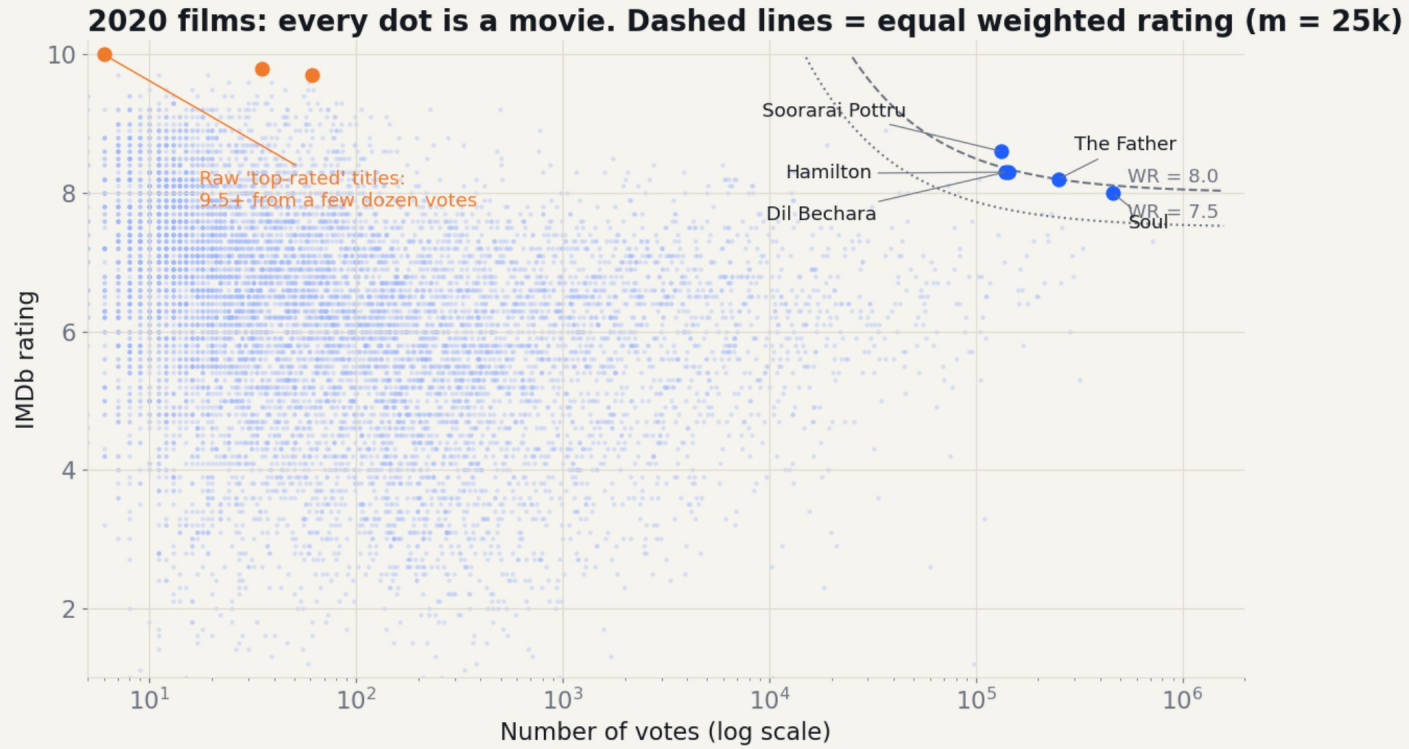
PART 1

What the data says about quality, stars and taste

The brief's three questions. Each one hides a measurement trap; how we avoid it is the lesson for the greenlight model.

Q1 · HIGHEST-RATED FILM OF 2020

Raw ratings crown films almost nobody watched



27 votes

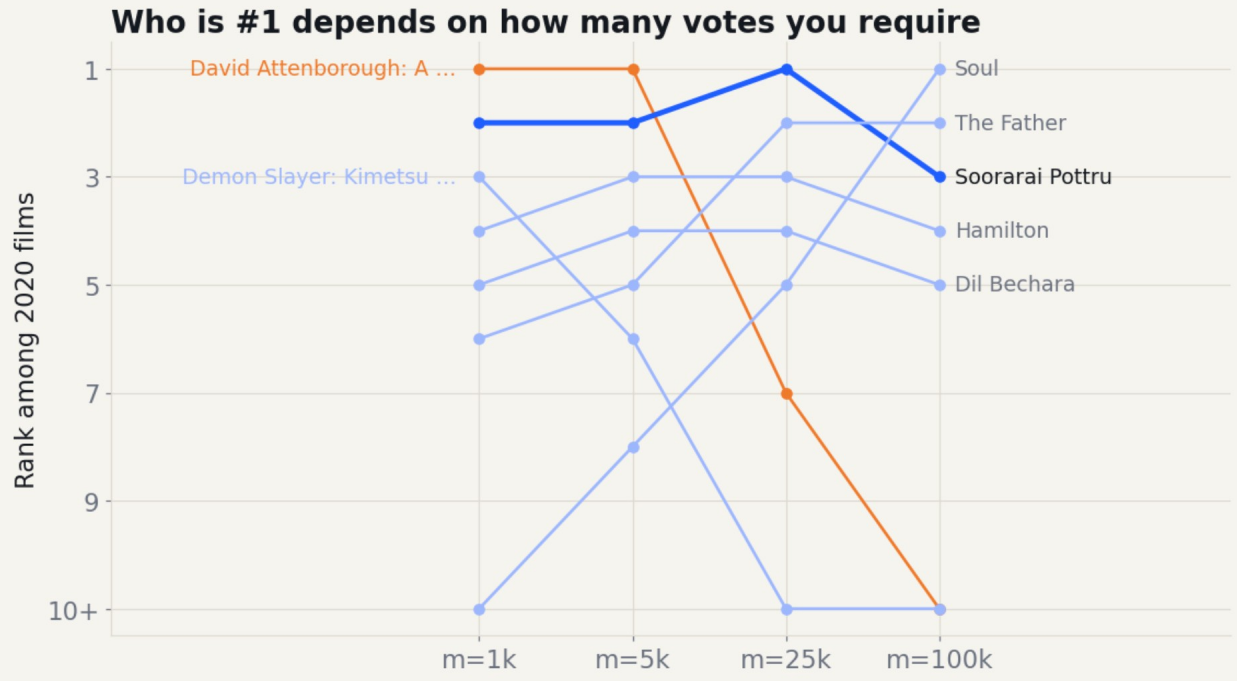
median for the 10 highest raw-rated 2020 films — mostly micro documentaries

WR = v/(v+m)·R + m/(v+m)·C

Bayesian shrinkage: a film earns its own score only as votes (v) pile up past m

Q1 · ANSWER

Soorarai Pottru leads 2020 — but #1 depends on the votes you require



Top 5, m = 25k	Rating	Votes	WR
Soorarai Pottru	8.6	133k	8.19
The Father	8.2	251k	8.00
Hamilton	8.3	143k	7.96
Dil Bechara	8.3	139k	7.95
Soul	8.0	460k	7.90

How to balance: choose m by the decision. Awards-style quality → m ≈ 25k. Mass appeal → m ≈ 100k, and **Soul** wins.

Q2 · MOST POPULAR ACTOR OF 2020

The obvious query names Tenet's background extras

Naive: IMDb "known for" titles

Person	2020 votes	Known for
Rob Purkiss	927k	Tenet; The Old Guard
Marisa Laurance	881k	Tenet; Borat 2
Richard George	809k	Tenet; The Midnight Sky
Alice Ford	806k	Tenet; Boss Level

"Known for" lists any credit, billed or not — so bit parts in one blockbuster dominate.

Fix: billing order, then define

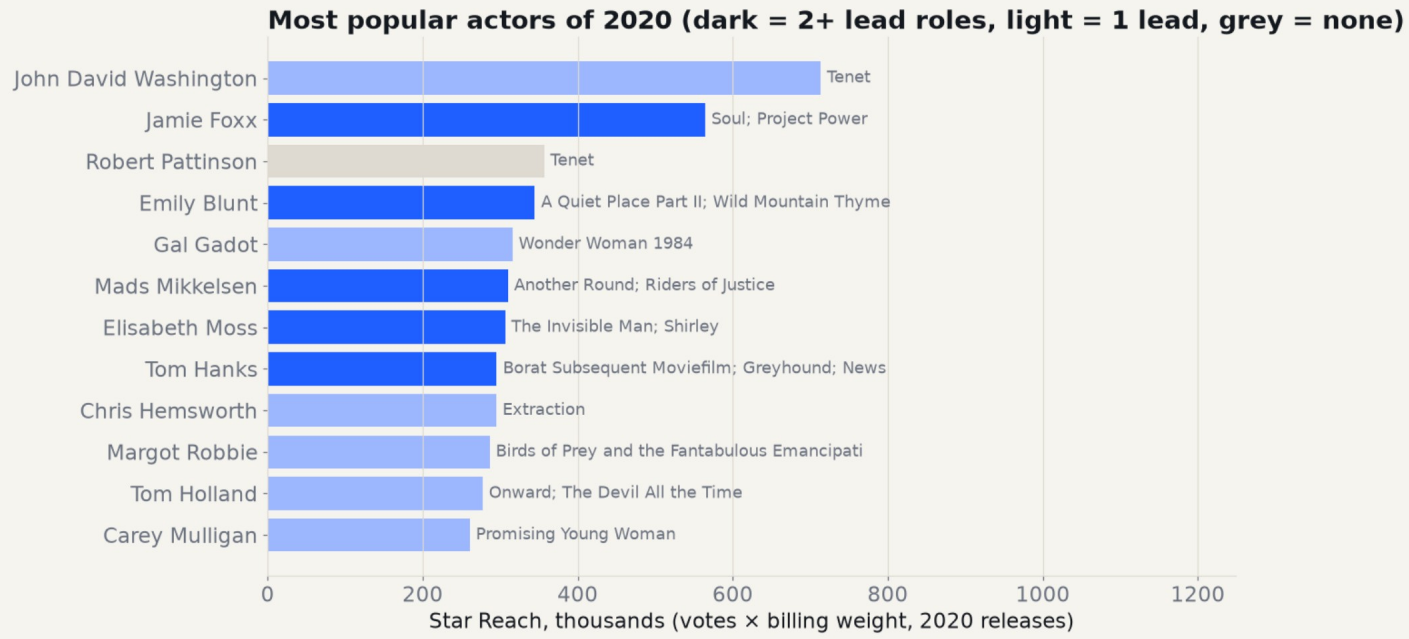
IMDb **title.principals**: credited cast of every 2020 film, in billing order, all countries. 78% agreement with an independent Wikipedia scrape.

Star Reach = Σ votes ÷ billing position
lead ×1 · 2nd ×½ · 3rd ×⅓ · 4th ×¼

Popularity = audience that saw you *carry* a film.

Q2 · ANSWER

John David Washington by reach; Jamie Foxx by reach and consistency



714k

John David Washington: one lead, in the year's most-voted film (Tenet)

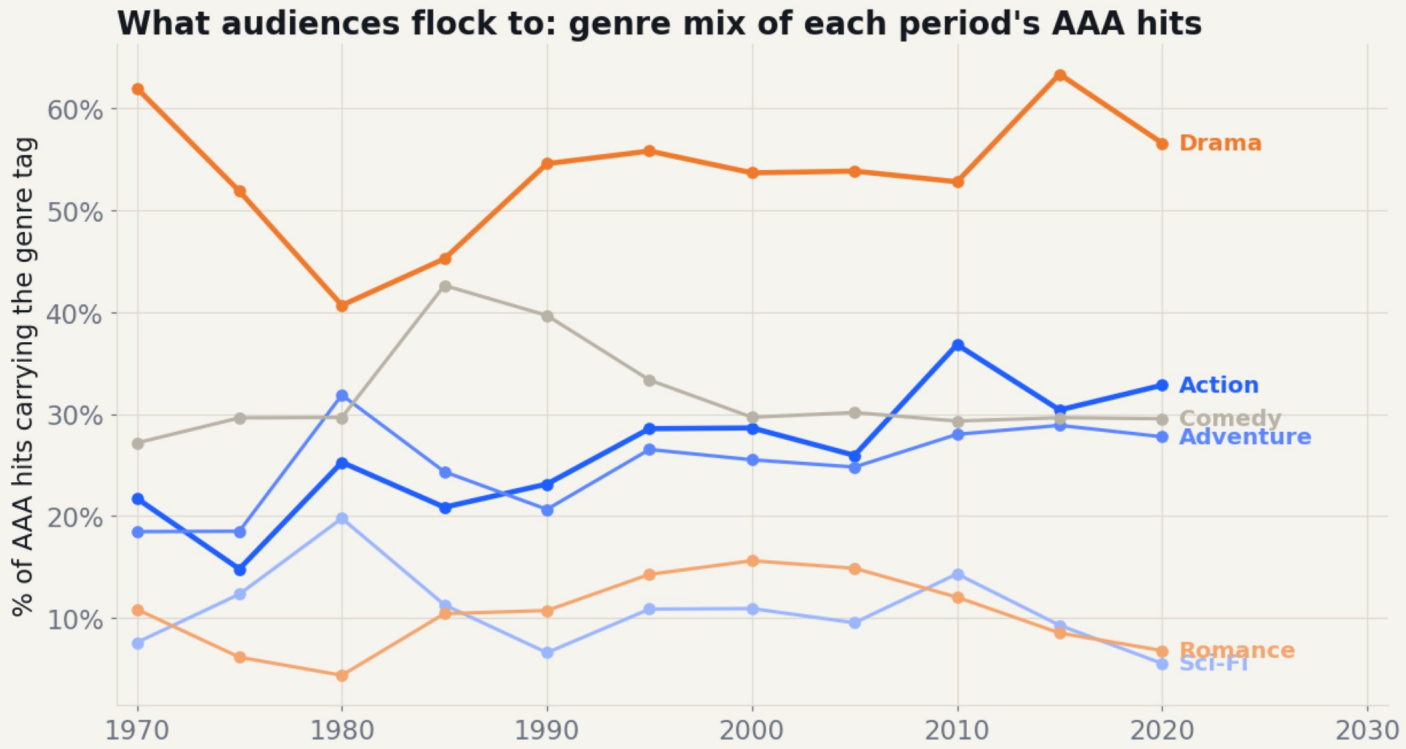
564k

Jamie Foxx: two leads (Soul, Project Power) and a 7.5 reach-weighted rating

Implication

Tenet's audience is arguably Nolan's. Price a single-hit star as a director bet.

Audiences moved from drama to genre-blended spectacle



3.5×

Adventure's over-representation among 2020–25 hits vs all releases. Animation 2.8×, Action 1.8×

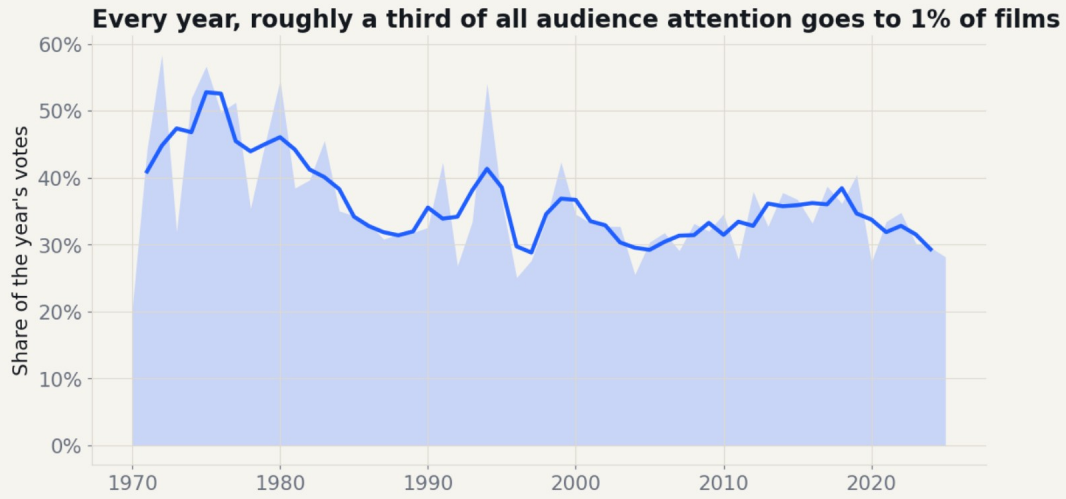
63% → 57%

Drama's share of hits, 2015–19 vs 2020–25, after decades as the backbone

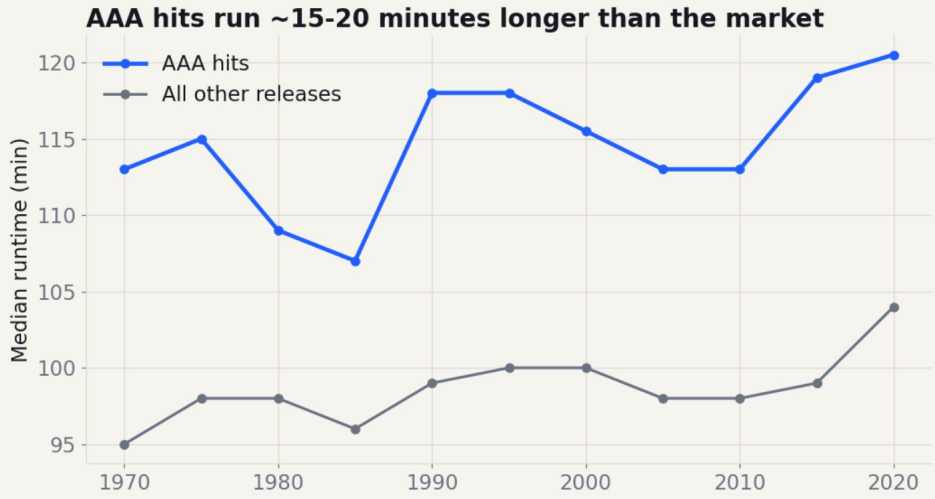
0.6×

Romance is under-represented among hits, and its share keeps shrinking

A hit-driven market: 1% of films take a third of all attention

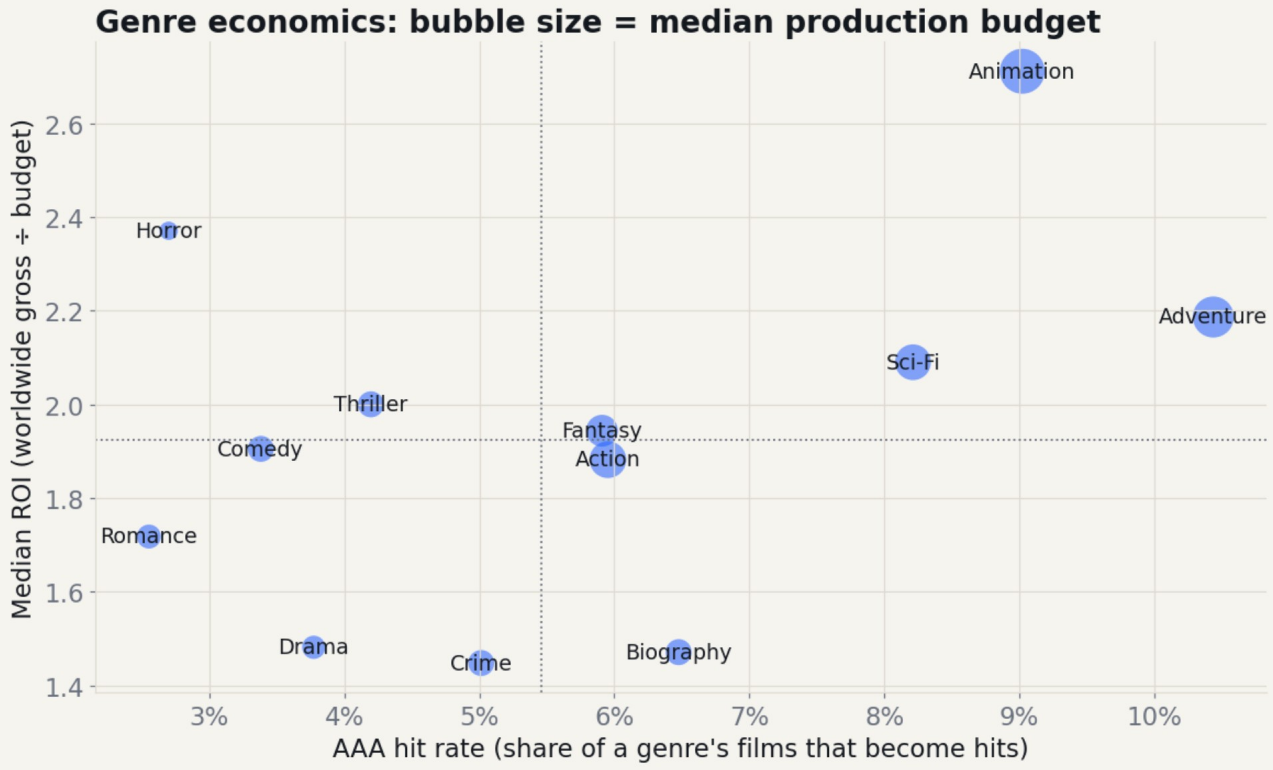


~33% of each year's votes go to the top 1% of films, steady since the mid-80s. **This is a power-law business.**



Hits run ~120 min vs ~104 for the market. Event films earn their length.

Animation wins on both axes; Horror is the ROI hedge



9% · 2.7×

Animation: highest hit rate and median ROI of any genre

2.4× on $10M

Horror: cheap, high-return, rarely a hit — ideal portfolio ballast

~1.5×

Drama, Crime and Biography: low hit rate and thin returns

PART 2

Defining a hit, then predicting it before greenlight

A metric the CFO accepts, a model the data scientists trust.

HIT METRIC

AAA Hit = top 5% audience reach in its year, with a quality gate

```
reach_pct(votes | release year) ≥ 0.95
AND IMDb rating ≥ 6.5
```

3.6% of films qualify — about the share a studio would call a tentpole success.

Reach is the money signal

Votes vs worldwide gross: Spearman **0.76**. Rating vs gross: only **0.22**.

Judged against its own year

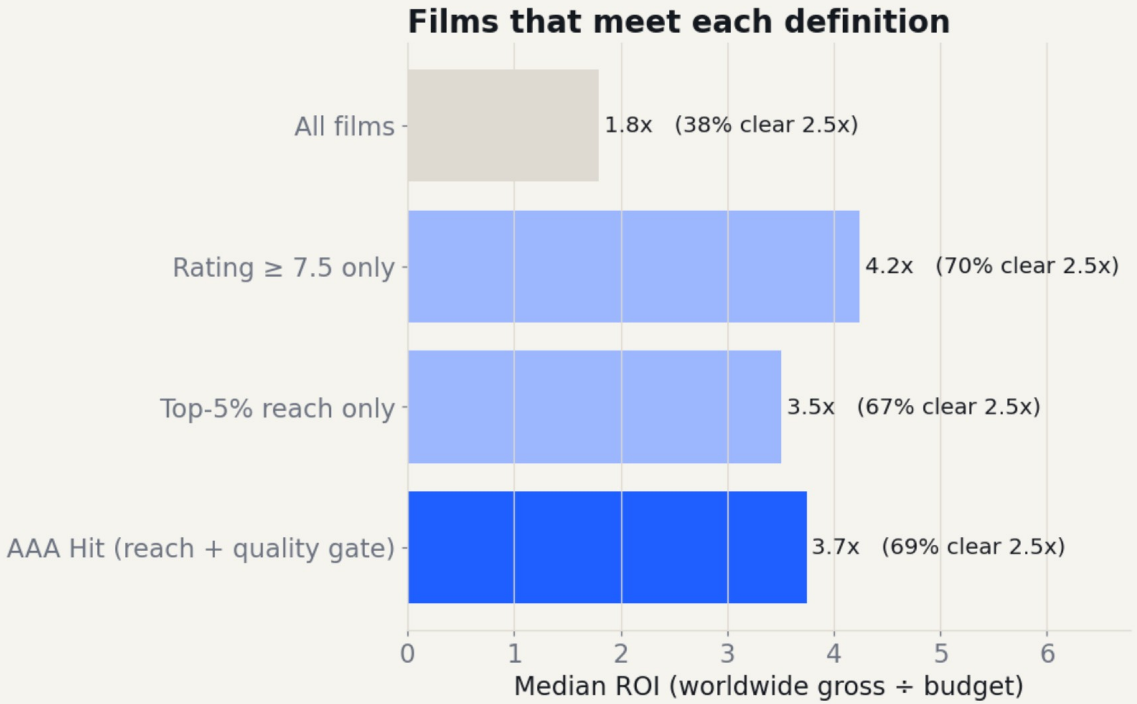
Percentiles within year remove IMDb's growth and the vote pile-up of old classics.

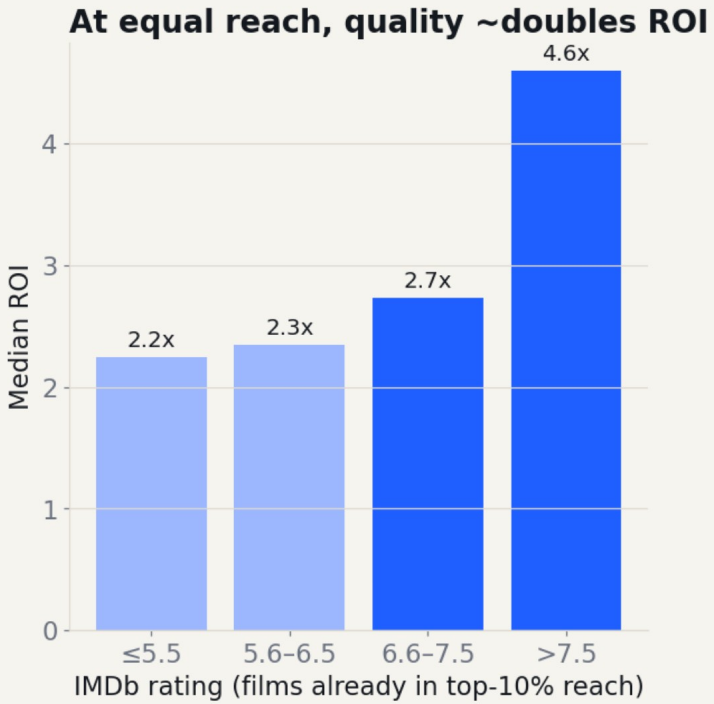
Quality protects the return

Among top-reach films, those rated >7.5 earn 4.6× ROI vs 2.2× for ≤5.5. The gate drops hate-watched films.

HIT METRIC · VALIDATION

Films meeting the definition return twice the market median





AAA hits: **3.7×** median ROI and **69%** clear 2.5× (theatrical break-even rule of thumb) vs 1.8× and 38% for all films. Rating-only finds efficient niche films (median gross $79M); the AAA definition keeps scale ($118M).

GREENLIGHT MODEL · DESIGN

Only information a studio has before it spends a dollar

Features known at greenlight

Track records of director, writer, top-3 cast and producers from *earlier years only* · franchise history · genre mix · planned runtime.

Out-of-time validation

Train 1975–2012 · tune 2013–16 · test on unseen 2017–2025. The studio uses it on future films, so we test that way.

Leakage audit: budget rejected

Budget exists mostly for famous films (hit rate 22% when known vs 1.3%). It leaks the outcome — and adding it makes test results slightly worse.

Decision metric

Precision at the top of the ranking. A studio funds a handful of projects; what matters is the hit rate of the ones we pick.

GREENLIGHT MODEL · RESULTS ON 2017–2025

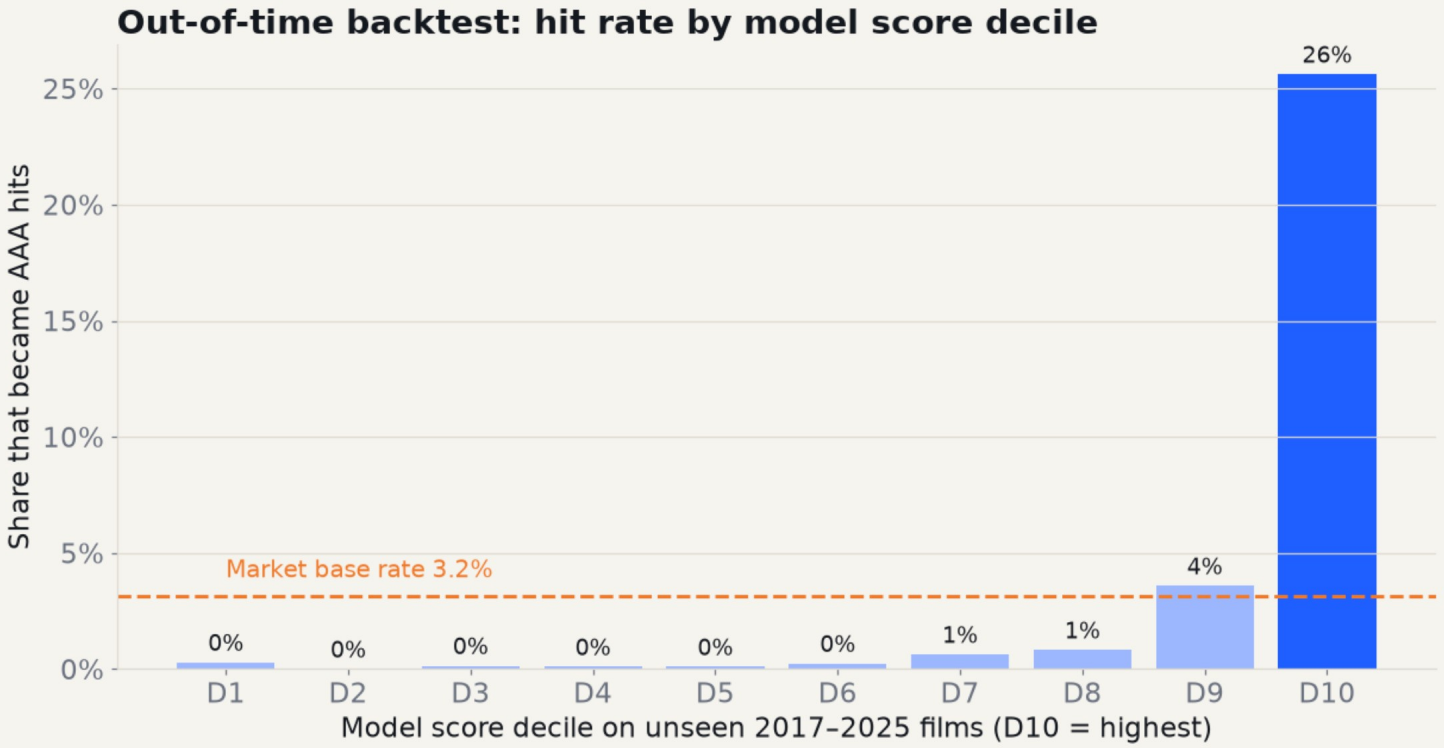
Talent history turns a weak model into a useful one

Model (test set, base rate 3.2%)	ROC-AUC	Hit rate, top 10%	Hit rate, top 2%
1 Logistic: genre + runtime	0.78	13%	19%
2 GBM: genre + runtime	0.80	14%	21%
3 + franchise / IP	0.81	15%	29%
4 + director & writer record	0.92	24%	60%
5 + cast & producer record — final	0.94	26%	61%
× Final + budget (rejected, leaks)	0.94	25%	58%

Genre "rules of thumb" alone beat chance 4×. Director & writer history is the step change; cast & producers add the last stretch. **61% of the top-2% scored films became AAA hits — 19× the market.**

VALUE REALIZATION · BACKTEST

93% of future hits sat in the model's top two deciles



26%

hit rate of the top decile vs 3.2% base

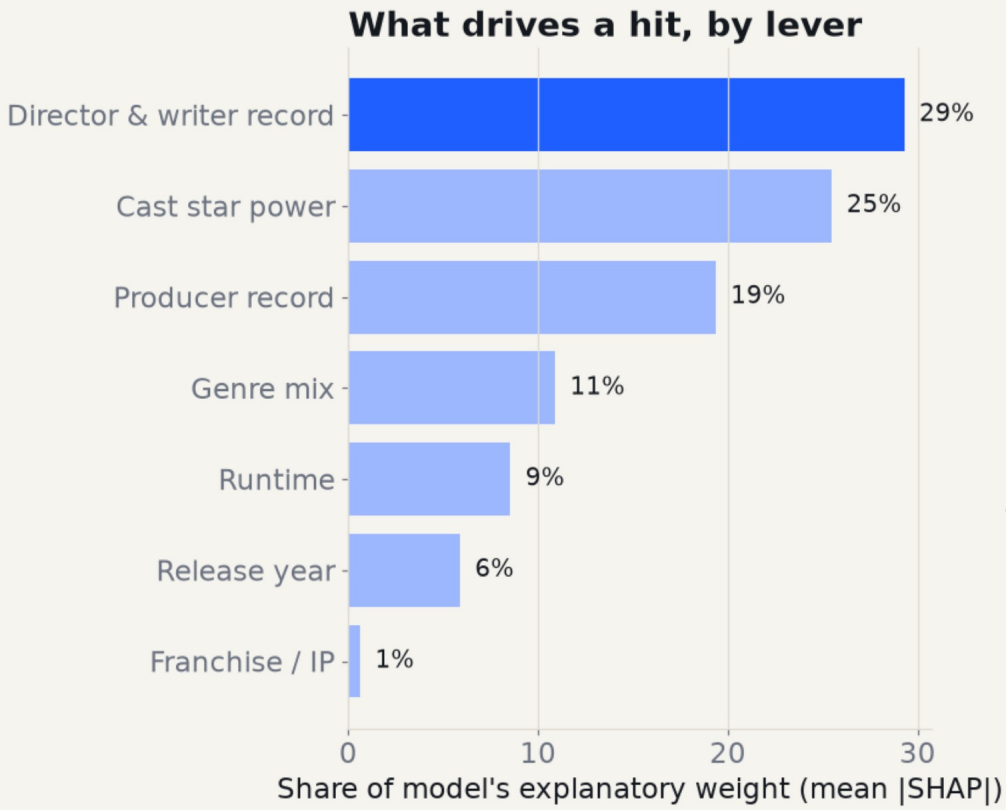
D1–D8: ≤1%

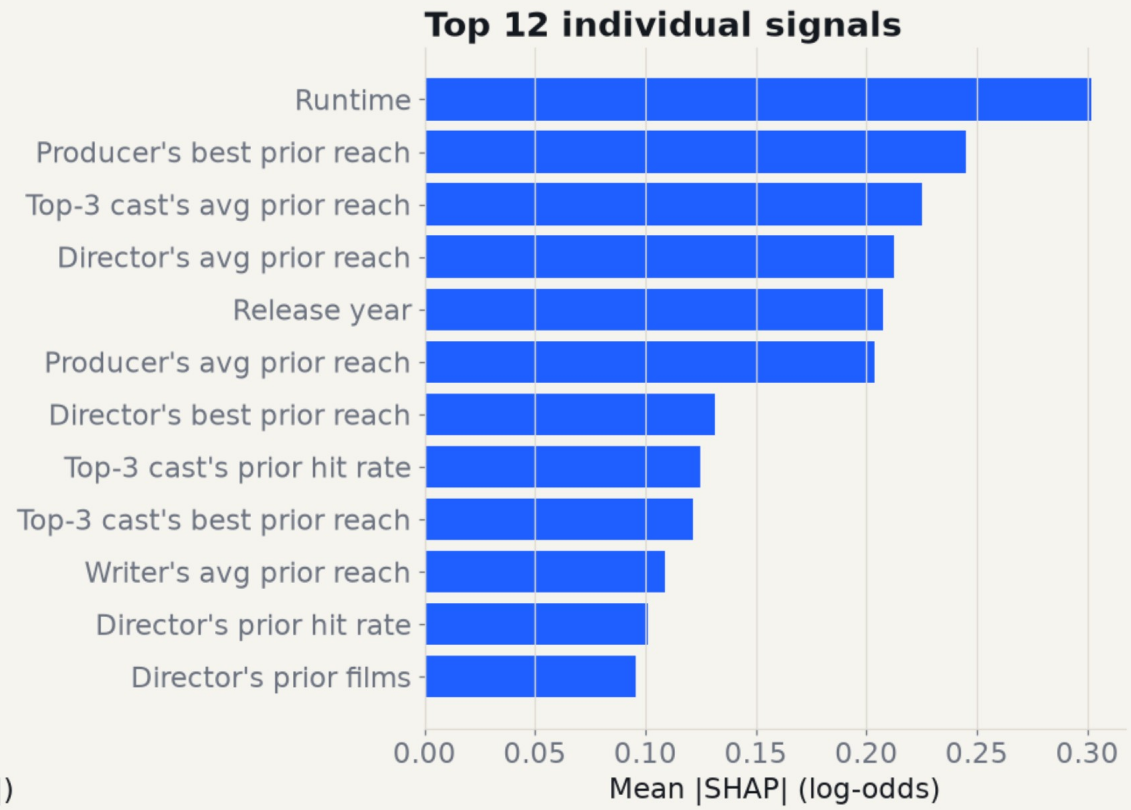
the bottom 80% produced almost no hits — a safe 'no' list

3.3× vs 1.8×

median ROI, top-quartile vs rest, among budgeted films 2013–19

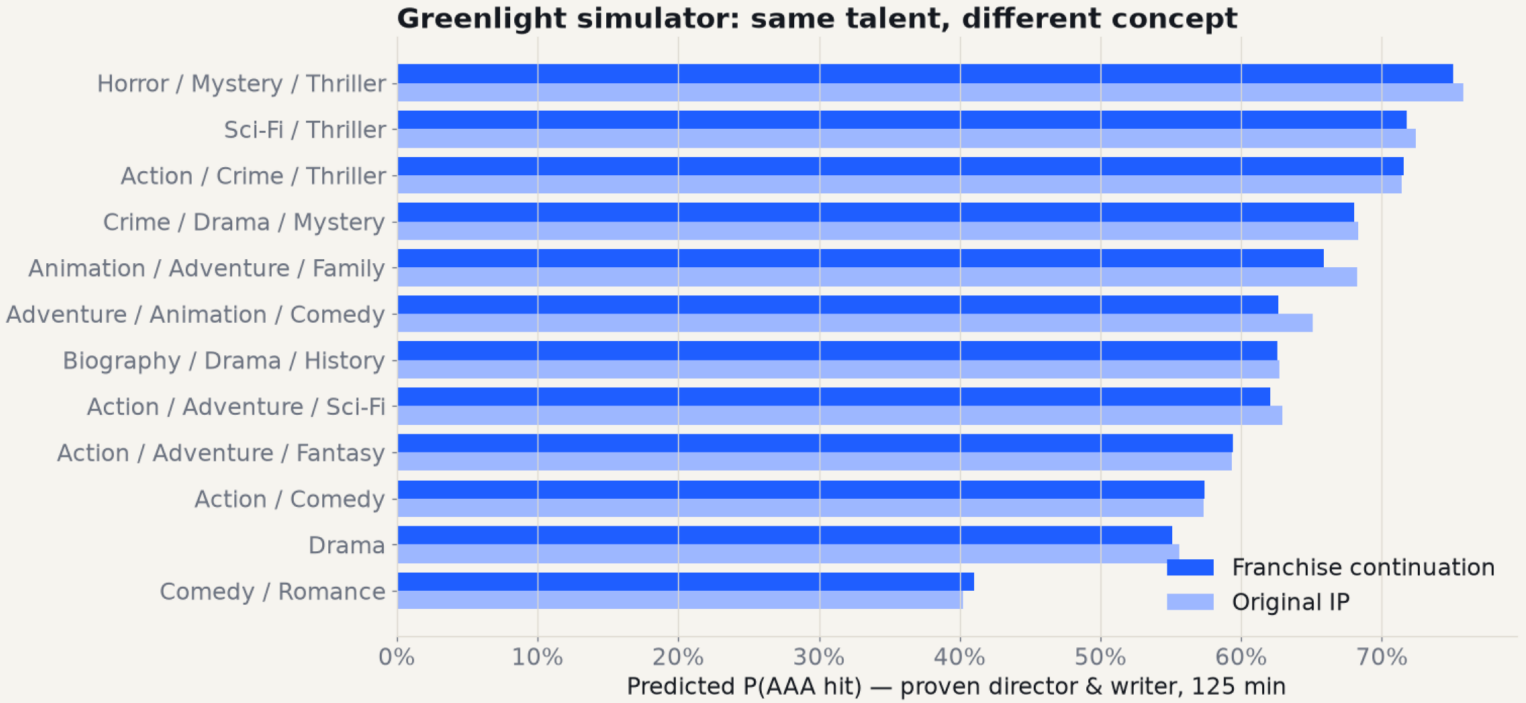
Who makes it matters far more than what it is





GREENLIGHT SIMULATOR

With proven talent, thriller blends score highest



76%

Horror / Mystery / Thriller, original IP, proven team, 125 min. Sci-Fi and Crime thrillers ~72%

~40%

Comedy / Romance with the same team

±2 pts

franchise continuation, once talent is proven

Scoring IMDb's unreleased 2026–28 slate: the model's watchlist

Upcoming film	Director	Top-billed cast	Year	P(hit)
Avengers: Doomsday	Russo brothers	Robert Downey Jr., Pedro Pascal	2026	89%
Dune: Part Three	Denis Villeneuve	Timothée Chalamet, Zendaya	2026	89%
Hunger Games: Sunrise on the Reaping	Francis Lawrence	Joseph Zada, Mckenna Grace	2026	87%
Werwulf	Robert Eggers	Aaron Taylor-Johnson, Willem Dafoe	2026	83%
Klara and the Sun	Taika Waititi	Jenna Ortega, Amy Adams	2026	74%
Incredibles 3	Peter Sohn	Samuel L. Jackson, Holly Hunter	2028	71%
Ally	Bong Joon Ho	Bradley Cooper, Dave Bautista	2027	58%

The franchise picks are the market's consensus. The signal is in the originals: **Werwulf**, **Klara and the Sun** and **Ally** score like franchise films because of who is making and starring in them. IMDb snapshot Oct 2026.

PART 3

The recommendation, and how we prove it paid off

RECOMMENDATION

The brief for our next AAA title

Lever	What the evidence says to do
Talent	Director, writer, producer and top-3 cast who have each delivered a top-reach film before (74% of what predicts a hit)
Concept	A thriller blend — horror-mystery, sci-fi or crime; or Animation / Adventure for family reach
Runtime	120–140 minutes: event-film scale
IP	Original IP is viable: franchise adds ~0 once talent is proven
Quality bar	Target ≥ 7.0 audience rating: quality roughly doubles ROI at equal reach
Portfolio	Pair the tentpole with 1–2 low-budget horror films as ROI ballast

Then score every candidate project with the model and fund only from the top two deciles.

VALUE REALIZATION

How we would prove the model is worth money

90-day shadow pilot

Score every pitch on the next greenlight slate **before** executives decide. Executives decide as usual; we log both. Releases then show whether the score would have changed the outcome.

KPI	Measure
Leading	Score of every pitch; share of slate from top-2 deciles
Lagging	Slate hit rate vs 3.2% base; median ROI vs 1.8×
Model health	Precision@top-decile on new releases; drift in talent features

BACKTEST, BUDGETED FILMS 2013–19

3.3×

median ROI of the top-scored quartile, vs 1.8× for the rest

On a $100M film, the median outcome moves from ~$185M to ~$330M gross.

Associational. The pilot tests it.

ROADMAP

From a case study to a product the studio uses every greenlight

Now

Data gaps

- Studio box-office & budget feed (public data ends ~2019)
- Fix fuzzy franchise matching
- De-duplicate multi-role cast credits

Next

Pre-release signals

- Trailer views, search trends, social buzz
- Streaming viewership (non-theatrical)
- Re-score projects as they progress

Later

Productionize

- dbt + Airflow on BigQuery / Snowflake
- Monthly retrain, model card, drift alerts
- Score API inside the greenlight workflow

RISKS AND LIMITATIONS

What this analysis does not prove

IMDb votes ≠ audience

Skews online, English-speaking, male. Validated against gross, but streaming hits are under-measured.

Money data is thin

9% of films, ends ~2019. Used to validate, never as a feature.

Correlation, not causation

The simulator shows where proven talent wins, not what a genre switch would cause.

Favours incumbents

Track records can't see a breakout newcomer. Pair the score with human scouting for new voices.

THE ASK

Approve a 90-day shadow pilot on the next greenlight slate.

No change to how decisions are made — just a score on every pitch, and a scorecard when the films release.

Code, SQL and model: shared repository · Questions welcome